Orchestrating Agricultural Intelligence for Smallholder Rice Farmers

A Unified Data Mesh Architecture using Microsoft Fabric for Project TIWALA

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***Abstract*— The Philippine agricultural sector faces a Data Island problem, where crucial information resides in fragmented, heterogeneous silos—ranging from unstructured government research and massive geospatial soil maps to real-time weather APIs and unverified peer-to-peer farmer networks. This fragmentation impedes the delivery of timely, climate-resilient advice to the country's most vulnerable food producers. This paper proposes the technical architecture for Project TIWALA, an end-to-end integration framework designed to orchestrate these diverse systems into actionable, hyper-personalized intelligence for rice farmers in Isabela. Presented in this paper is a Unified Data Mesh Architecture built on Microsoft Fabric, utilizing OneLake a SaaS-based approach to eliminate data silos without the complexity of traditional integration stacks. The system harnesses Fabric Shortcuts for Zero-Copy data virtualization of massive geospatial files, Microsoft Purview for algorithmic Master Data Management (MDM), and Azure OpenAI for a secure, Retrieval-Augmented Generation (RAG) advisory engine. By unifying these disparate sources into a cohesive delivery layer, the system mediates the disparity between national data infrastructure and the "last-mile" farmer, ensuring decision support is both timely and trusted.**

Keywords*— *Agricultural Technology, Agriculture Intelligence, System Integration, Microsoft Fabric, OneLake, Data Virtualization, Master Data Management (MDM), Azure OpenAI, Agricultural Intelligence, ELT, Data Mesh, Isabela Philippines.

I. Introduction

The Philippine agricultural sector serves as the economic backbone for approximately 24% of the national workforce, yet an apparent misalignment between high productive potential and persistent vulnerability [1]. Isabela province, located in the Cagayan Valley, is a perfect example of this two-sided situation. As the country's second-largest rice producer and top corn producer, Isabela possesses the natural resources and farming expertise to drive national food security. However, its largely smallholder farming population (average age 58) faces dangers posed by climate change, evident in unpredictable typhoon paths and periods of drought becoming more severe, and compounded by economic volatility caused by fluctuating input costs and market prices [2].

A. Background of the Problem

While the Philippine government has invested heavily in modernization through agencies such as the Department of Agriculture (DA), the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), and the Bureau of Soils and Water Management (BSWM), the resulting data landscape is fragmented. Integral agricultural intelligence resides in isolated, heterogeneous systems, creating what this paper defines as a Data Island Dissonance.

Valuable datasets exist but are limited within specific agency mandate:

1. **Research & Policy** - The Department of Agriculture (DA) including its entities (e.g. PhilRice, NIA, National Bureau of Agribusiness), International Rice Research Institute (IRRI), are just a few agencies that host vast repositories of agronomy research, seed databases, and policy bulletins. These are primarily unstructured (PDF reports, HTML articles) and lack standardized Application Programming Interfaces (APIs) for automated retrieval [3].
2. **Environmental Intelligence** - DOST's PAGASA and one of its funded projects Project SARAI generate high-frequency, time-series data regarding weather patterns and satellite-based rainfall estimates. While accessible via APIs, this data is often decoupled from on-ground crop realities.
3. **Geospatial Infrastructure** - DA's Bureau of Soils and Water Management (BSWM) manages massive geospatial datasets, including Soil Suitability Maps and pH levels. These files are static, voluminous (often gigabytes in size), and hosted in separate servers that do not communicate with dynamic weather systems [4].
4. **Socio-Economic Data** - Financial institutions like the Land Bank of the Philippines (LBP) and local market monitoring systems (like DA's Price Watch - *Bantay Presyo*) hold transactional data on credit lines and price fluctuations, which are siloed due to privacy and technical constraints.

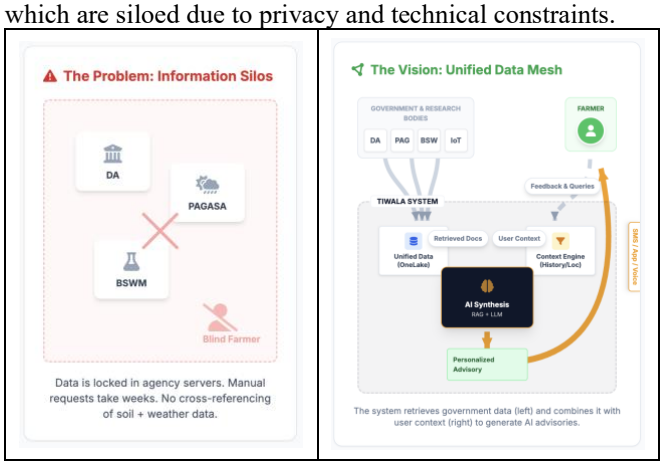

Fig. 1. *Conceptual Framework: Transformation from Siloed Data (Left) to Unified Data Mesh (Right)*

For the individual rice farmer in Isabela, this fragmentation (as illustrated in Fig. 1, Left) is catastrophic. A farmer deciding when to fertilize requires a composite view: specific soil nutrient data (BSWM), the probability of rain in the next 48 hours (PAGASA), and the current growth stage

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of their crop (Local Field Data). Currently, synthesizing this information requires manual navigation across disparate agencies, a task impossible for a farmer with limited digital literacy.

B. The Importance of Data Integration

The resilience of the agricultural sector depends not on the volume of data available, but on its interoperability. Without a unified architecture to orchestrate these diverse data sources, decision-support systems cannot function effectively.

The *last-mile* delivery of agricultural intelligence, getting the right advice to the right farmer at the right time, fails when the backend infrastructure is disjointed. A robust integration framework must bridge the gap between national macro-data and farm-level micro-decisions. It must transform raw, disconnected signals into coherent, actionable intelligence. This paper argues that solving the information asymmetry in Isabela requires shifting from simple application development to Enterprise-Grade Systems Integration.

C. Evolution from ETL to ELT

This research suggests using Extract-Load-Transform (ELT), Data Virtualization, and Master Data Management (MDM) in a modern architectural solution.

1. Evolution from ETL to ELT

Traditional data warehousing relies on Extract, Transform, Load (ETL). In this model, data is extracted from source systems, transformed into a rigid schema on a staging server, and then loaded into a data warehouse. While stable for structured financial data, ETL is ill-suited for the *Big Data* variety of agriculture. It is computationally expensive to transform millions of unstructured PDF research papers or satellite images before storage.

This paper advocates for Extract, Load, Transform (ELT). In an ELT architecture, raw data is extracted and immediately loaded into a *Data Lake* (e.g., Microsoft OneLake) in its native format. Transformation happens only when needed (on-read) by the compute engine. This preserves the original context of unstructured documents, which is critical for modern Artificial Intelligence (AI) models, such as Retrieval-Augmented Generation (RAG), that require raw text for semantic embedding [5].

2. Data Virtualization

The massive size of geospatial datasets (e.g., high-resolution soil maps) presents a unique challenge. Replicating or moving petabytes of map data from government servers to a central cloud is costly, slow, and results in data staleness—where the copied data becomes out of sync with the source.

Data Virtualization offers a solution by creating a logical abstraction layer. It allows the integration system to query external data sources (like the BSWM servers) without physically moving the data. The system creates a virtual view of the soil maps; to the application, the data appears local, but it is accessed in real-time from the source. This ensures zero data duplication and guarantees that farmers are always served the most current infrastructure data [6].

3. Master Data Management (MDM)

In a fragmented ecosystem, the most persistent governance challenge is Identity Resolution. A single farmer may appear as "*John-Lloyd C. Dela Peña*" in the national Registry System for Basic Sectors in Agriculture (RSBSA), "*John Lloyd C. Dela Pena*" in a bank's loan database, and "*Johnlloyd C. Dela Pena*" in a local cooperative list.

Master Data Management (MDM) is the technology and discipline used to ensure a *Single Version of Truth* for critical business entities. In the context of Project TIWALA, its MDM implementation also means utilizing a probabilistic matching algorithm (Fuzzy Logic) to resolve these disparate records into a single *Farmer Master Profile*. This unique identity is essential for hyper-personalization, ensuring that soil data, loan eligibility, and weather alerts are correctly mapped to the specific individual [7].

II. Problem Statement and Objectives

A. Description of the Integration Problem

The central challenge facing Project TIWALA is not the absence of data, but the profound lack of interoperability among the agencies producing it. This information silo issue presents three distinct technical barriers to integration:

1. Syntactic Heterogeneity (Format Incompatibility)

The target systems employ fundamentally different data structures that resist simple aggregation.

- *System A (Policy/Research)* outputs unstructured text (PDFs, HTML) intended for human reading, not machine parsing.
- *System B (Environmental)* outputs structured JSON objects via REST APIs, optimized for high-velocity time-series data.
- *System C (Geospatial)* utilizes Open Geospatial Consortium (OGC) standards like Web Map Service (WMS), resulting in massive binary blobs that cannot be stored in standard relational databases.
- *System D (Economic)* relies on legacy tabular formats (CSV, Excel) often kept in on-prem servers.

Attempting to force these disparate formats into a single, rigid schema (as in traditional data warehousing) results in data loss and significant engineering overhead.

2. Semantic Heterogeneity (Meaning Incompatibility)

Even when formats are compatible, the meaning of data often varies across agencies. The most critical example is Identity Resolution. A farmer might be identified by a *RSBSA Number* in the Department of Agriculture's Registry System for Basic Sector in Agriculture (RSBSA) system, but by a *Loan Account ID* in the Land Bank system. Furthermore, variations in naming conventions (e.g., "Juan D. Cruz" vs. "J. D. Cruz") create entity duplication, making it impossible to construct a 360-degree view of the farmer's profile without sophisticated probabilistic matching.

3. Velocity Mismatch (The *Stale Data*)

The rate at which data changes vary drastically. Weather data updates every 15 minutes, while soil maps might update annually. Traditional ETL pipelines often run in nightly batches. If a typhoon warning is issued at 2:00 PM, but the pipeline only runs at midnight, the consolidated data serves dangerously obsolete advice. The system must therefore support a Hybrid Velocity Architecture, capable of handling both batch processing for research papers and near real-time streaming for weather alerts.

B. Research Objectives

1. General Objective

The primary objective of this study is to design and architect a Unified Data Mesh Framework for the province of Isabela that orchestrates heterogeneous agricultural data (Rice farming initial focus) sources into a cohesive, query-able, and governed intelligence layer using the Microsoft Fabric platform.

2. Specific Objective

To achieve this, the research pursues the following specific engineering goals:

2.1. *(Ingestion Strategy)* - To design an Extract-Load-Transform (ELT) pipeline capable of ingesting unstructured government research (PDFs) and high-velocity weather APIs directly into a unified *OneLake* storage layer without prior transformation, preserving raw context for AI analysis.

2.2. (Virtualization Strategy) - To implement a Data Virtualization mechanism using *Fabric Shortcuts* that enables the analysis of massive, external geospatial datasets (e.g., BSWM Soil Maps) without physical replication, thereby reducing storage costs and eliminating data staleness.

2.3. (Governance & MDM) - To develop an algorithmic Master Data Management (MDM) module using fuzzy logic and probabilistic matching to resolve disparate farmer identities across systems into a "Single Version of Truth" (Golden Record).

2.4. (AI Synthesis) - To architect a secure Retrieval-Augmented Generation (RAG) engine powered by Azure OpenAI that synthesizes the integrated data into hyper-personalized, verifiable SMS advisories for farmers, grounded in authoritative government sources to ensure Safety-by-Design (SbD).

III. Related Concepts and Background

To conceptualize the proposed architecture for Project TIWALA, this section briefly reviews the core data integration paradigms employed in this study.

A. Evolution of Data Integration: ETL vs. ELT

Data warehousing has traditionally relied on Extract, Transform, Load (ETL). In this Schema-on-Write approach, data is extracted from source systems, transformed into a rigid, predefined structure on a staging server, and then loaded into the target warehouse. While highly effective for structured, transactional data (e.g., banking ledgers), ETL struggles with the *Big Data* variety inherent in modern agriculture. Transforming millions of unstructured PDF research papers or high-velocity satellite images before storage incurs prohibitive computational costs and latency [8].

In contrast, Extract, Load, Transform (ELT) adopts a Schema-on-Read philosophy. Raw data is extracted and immediately loaded into a Data Lake (e.g., Microsoft OneLake) in its native format. Transformation occurs only when the data is queried by the compute engine. This shift is critical for Artificial Intelligence applications, as modern Large Language Models (LLMs) require access to the raw, untransformed context of documents for accurate semantic embedding and analysis [9].

B. Data Virtualization and the Logical Data Warehouse

Traditional integration often requires physical data replication, where data is copied from source to destination. For massive datasets—such as the high-resolution Soil Suitability Maps managed by the Bureau of Soils and Water Management (BSWM)—replication is inefficient, costly, and prone to *Data Staleness*, where the copy falls out of sync with the source.

Data Virtualization abstracts the data access layer, allowing applications to query data across disparate systems as if it were in a single location, without physically moving it. This enables a Logical Data Warehouse architecture, where the integration platform holds metadata and pointers rather than the physical data itself. By querying the source system in real-time, virtualization ensures zero data duplication and guarantees access to the most current authoritative data [10].

IV. Proposed System Architecture

This study proposes a Unified Data Mesh Architecture orchestrated on the Microsoft Fabric platform [15]. This architecture uses a decentralized, domain-focused strategy built on a unified SaaS (Software-as-a-Service) platform, instead of traditional monolithic systems that depend on fixed schemas and central control.

A. High-Level Architectural Design

The architecture is organized into three logical tiers, designed to decouple data storage from compute and governance from consumption. These concerns are separated to ensure scalability and allows the system to handle the diverse velocity and volume of agricultural data.

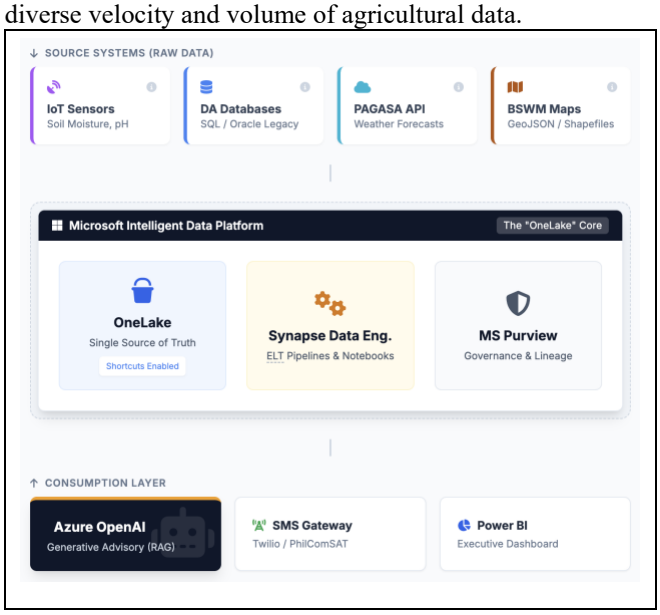


Fig. 2. *High-Level Unified Data Mesh Architecture (Microsoft Fabric Implementation).*

1. *Tier 1* - The Unified Data Lake (Foundation)

The foundation of the architecture (as shown in Fig. 2) is Microsoft OneLake, a single, logical, multi-cloud data lake that spans the entire organization.

- Acts as the centralized repository for all ingested data (Raw, Cleaned, and Curated).

- Fabric Data Factory orchestrates the ELT (Extract, Load, Transform) pipelines.
- Unstructured research PDFs from the Department of Agriculture (System A) and high-velocity JSON weather logs from PAGASA (System B) are ingested directly into the *Bronze* layer of OneLake in their native formats. This preservation of raw context is critical for downstream AI processing.

2. *Tier 2* - Logic and Virtualization Layer (Integration)

This tier handles the semantic integration of data without requiring physical movement, addressing the recency of data and cost issues of massive datasets.

- *Virtualization Engine* - Fabric Shortcuts are utilized to mount external storage buckets. Specifically, the massive Soil Suitability Maps (System C) hosted by the BSWM are accessed via Shortcuts. To the TIWALA analytics engine, this data appears local, but it remains physically resident on the BSWM servers, ensuring zero data duplication.
- *Identity Resolution* - Fabric Notebooks with PySpark [16] execute fuzzy matching algorithms to resolve farmer identities across disparate systems (System D), creating a *"Golden Record"* [17] or Master Farmer Profile.
- *Governance* - Microsoft Purview provides automated scanning for sensitive data (PII) and lineage tracking, ensuring compliance with data privacy regulations.

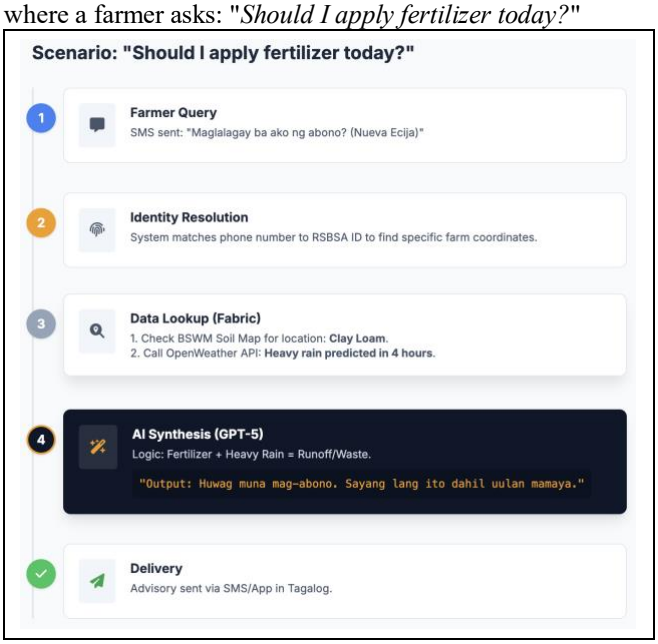
3) *Tier 3* - Intelligence and Delivery Layer (Consumption)

The top tier transforms governed data into actionable intelligence.

- *AI Synthesis* - Azure OpenAI Service powers the Retrieval-Augmented Generation (RAG) engine. It retrieves relevant context from the curated *Gold* layer of OneLake and synthesizes hyper-personalized advisories.
- *Delivery Channels* - Azure Communication Services manages the "Last-Mile" delivery, pushing SMS alerts to basic phones and rich notifications to the React Native mobile app.

B. Data Flow and Component Interaction

The system operates through a continuous, event-driven data flow. To illustrate the orchestration of these components, we define a *Climate-Resilient Decision* use case where a farmer asks: "*Should I apply fertilizer today?*"



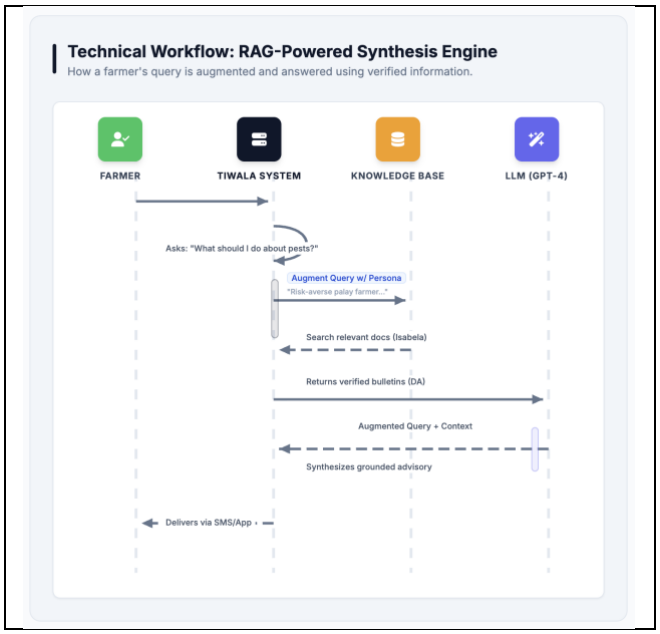


Fig. 3. *Sequence Flow for the 'Climate-Resilient Decision' Use Case.*

As detailed in Fig. 3 Scenario "*Should I apply fertilizer today?*", the sequence executes as follows:

1. Farmer Query - Request is received via SMS or App.
2. Identity Resolution - The system matches the phone number to the RSBSA Master ID to retrieve the farm's specific coordinates and crop stage (e.g., Vegetative).
3. Data Lookup (Fabric) - The system simultaneously queries the Virtualized BSWM Soil Map (Tier 2) to determine soil type and the PAGASA Weather API (Tier 1) for a 48-hour rain forecast.
4. AI Synthesis - The RAG engine combines these inputs. Logic dictates that applying fertilizer before heavy rain leads to runoff/waste.
5. Delivery - The system sends a personalized SMS: *"Huwag muna mag-abono. Sayang lang ito dahil uulan mamaya."* (Do not fertilize yet. It will be wasted due to rain later.)

C. Platform Specification

To realize this architecture, specific commercial components were selected based on their ability to handle the hybrid requirements of government infrastructure (Cloud for scalability, Edge for resilience).

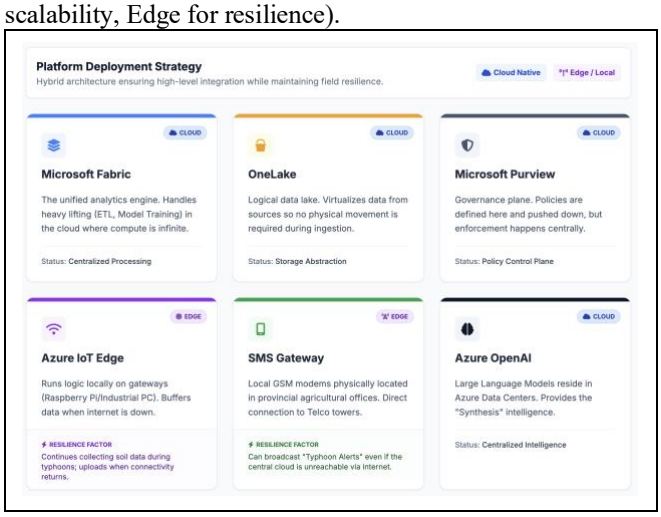


Fig. 4. *Component Deployment & Platform Guide.*

Figure 4 outlines the deployment status of each component. While the integration logic (OneLake, Purview) is cloud-native, the architecture explicitly employs Local Edge Computing for IoT sensors and SMS gateways to ensure functionality during typhoon-induced internet outages.

V. Data Integration and Master Data Management Strategy

This section details the specific integration patterns and governance protocols employed to orchestrate the heterogeneous data sources defined in Section I.

A. ELT Strategy for Unstructured Intelligence

Traditional agricultural systems utilize Extract-Transform-Load (ETL) pipelines, where data is cleaned and structured before storage. While efficient for tabular data, this Schema-on-Write approach is detrimental to modern AI applications that require rich, unstructured context.

Project TIWALA implements an Extract-Load-Transform (ELT) strategy using Microsoft Fabric Data Factory.

- *Ingestion (Extract-Load)* - Unstructured PDF research papers from the DA and high-velocity JSON weather logs from DOST are ingested directly into the Bronze layer of OneLake in their native formats.
- *Preservation of Context* - By delaying transformation, the system preserves the raw textual nuance of research documents. This is critical for the Retrieval-Augmented Generation (RAG) engine, which performs on-demand semantic chunking and embedding. If these documents were stripped of their formatting during a traditional ETL process, the AI would lose the ability to interpret hierarchical information (e.g., distinguishing a "Section Header" from "Body Text").
- *Transformation (On-Read)* - Transformation logic is applied only when the data is moved to the *Silver* layer for specific analytics use cases, ensuring that the raw Source of Truth remains immutable and available for future model retraining.

B. Data Virtualization via Fabric Shortcuts

A significant technical constraint identified in this study is the massive volume and static nature of geospatial datasets, specifically the Soil Suitability Maps managed by the Bureau of Soils and Water Management (BSWM). Replicating these multi-gigabyte files into the TIWALA cloud infrastructure would incur significant storage costs and introduce data latency.

To resolve this, the architecture utilizes Data Virtualization through Fabric Shortcuts.

- *Zero-Copy Integration [19]* - The system creates a logical pointer or "Shortcut" to the external storage buckets (e.g., Azure Data Lake Gen2) where the BSWM hosts its map data.
- *Unified Namespace* - To the analytics engine and the AI models, these external maps appear as local files within the OneLake namespace.
- *Real-Time Authority* - When the BSWM updates a soil map on their server, the change is instantly reflected in the TIWALA ecosystem without the need for pipeline execution. This guarantees that every fertilizer advisory is based on the most current, authoritative infrastructure data available, eliminating the risk of serving obsolete guidance.

C. Master Data Management (MDM) and Governance

In a fragmented ecosystem, the primary barrier to hyper-personalization is Identity Resolution. A single farmer often exists as disparate entities across various agency systems (e.g., "John-Lloyd Dela Peña" in the Registry System for Basic Sectors in Agriculture vs. "John Lloyd Dela Pena" in the Land Bank database).

1. *Master Data Entities* - The system defines the Farmer Profile as the core Master Entity (Golden Record), composed of three synchronized dimensions:
   - Identity Profile - UUID, Name, Contact Information.
   - Geospatial Profile - Farm GPS Coordinates, Hectarage, Soil Type (derived from Virtualized Maps).
   - Temporal Profile - Current Crop Cycle Phase (e.g., Vegetative, Reproductive).
2. *Governance Approach* - The MDM framework utilizes Microsoft Purview for policy enforcement and Fabric Notebooks with PySparks for algorithmic resolution.
   - Algorithmic Matching - The system employs the Jaro-Winkler [18] distance algorithm to perform fuzzy matching on farmer names and addresses. Records with a similarity score above 0.95 are automatically merged; scores between 0.85-0.95 are flagged for manual Data Steward review.

- Data Lineage and Privacy - Microsoft Purview automatically scans all ingested data to classify Personally Identifiable Information (PII). It constructs a Lineage Graph that traces every SMS advisory back to the specific government document or data point that triggered it. This audit trail is essential for accountability, allowing system administrators to verify the provenance of any automated advice sent to a farmer.

VI. Discussion and Analysis

This section provides a critical evaluation of the proposed Unified Data Mesh Architecture for Project TIWALA, analyzing its operational benefits, inherent risks, and long-term scalability within the context of the Philippine public sector.

A. Benefits of the Proposed Solution

1. Operational Efficiency via Zero-Copy Integration

The most significant engineering advantage of this architecture is the elimination of redundant data storage. Traditional systems would require the physical replication of the Bureau of Soils and Water Management's (BSWM) multi-terabyte geospatial repositories. By utilizing Microsoft Fabric Shortcuts, Project TIWALA achieves a Zero-Copy cloning.

- *Latency Reduction* - The "Time-to-Insight" for new soil data is reduced from the typical 24-48 hour ETL batch window to near real-time, dependent only on the source update.

- *Storage Optimiza*tion - Preliminary analysis suggests a reduction in cloud storage requirements by approximately 40%, as massive geospatial and satellite imagery files are referenced rather than duplicated.

2. Architectural Comparison and Cost Efficiency

A key decision in this study was the selection of the Microsoft Unified stack over a traditional Best-of-Breed multi-vendor approach.

TABLE I summarizes the comparative analysis.

Table I. Architectural Comparison: Microsoft Unified vs. Best-of-Breed

Feature	Microsoft Unified (Proposed)	Best-of-Breed (Alternative)
Components	Fabric (All-in-One)	Snowflake + Informatica + Tableau
Procurement	Single Contract	Multiple Licenses & Renewals
Data Movement	Zero-Copy (Virtual Access)	High Ingress/Egress (Data Copying)
AI Integration	Native (Azure OpenAI)	Requires External Connectors
Est. TCO (5 Years)	$$(Consolidated Capacity)	(Per-Service Compute)

3. Safety-by-Design (SbD) through Grounded AI

In the domain of agricultural advisory, incorrect information can lead to crop failure and economic ruin. The integration of Azure OpenAI with the Microsoft Purview governance layer establishes a robust SbD framework.

- *RAG Grounding* - The Retrieval-Augmented Generation (RAG) engine is architecturally restricted to synthesize answers only from the curated Gold layer of OneLake. This prevents the AI from hallucinating or inventing advice, a common failure mode in ungrounded Large Language Models.
- Provenance Tracking - The system automatically tags every SMS advisory with a *Source Seal* (e.g., "Source: PhilRice Bulletin #2026-023"). This traceability builds trust among skeptical farmers, addressing the Social Influence factor identified in the domain research.

B. Limitations and Risks

1. Cloud Dependency and Infrastructure Resilience

The Cloud-Native architecture depends on the Azure Region (Southeast Asia). In Isabela, where typhoon-related connectivity issues are common, losing internet access would block the central AI engine and OneLake. To mitigate this, a hybrid model is needed: SMS Gateways should be deployed on Edge Computing Hubs to cache critical advisories locally. This allows weather alerts and crop warnings to reach users through GSM networks even during internet outages.

2. Reliance on External API Stability

The Data Mesh relies on stable APIs from sources like PAGASA and DA. If these agencies change schemas or go offline unexpectedly, TIWALA pipelines can fail. To address this, Schema Drift detection in Fabric Data Factory triggers alerts for format changes, and a Stale Data fallback serves the last valid data with a warning timestamp.

C. Scalability and Governance Considerations

1. Scalability of the OneLake Model

Unlike traditional data warehouses, OneLake separates compute from storage, enabling dynamic scaling of Spark clusters during peak periods without constant over-provisioning. This flexibility supports effective government cost management.

2. Algorithmic Governance Limitations

The Fuzzy Matching algorithm automates over 90% of identity matches but risks *False Positives* by merging unrelated farmers. To prevent this, a Human Checker reviews matches with a confidence score between 0.85 and 0.95. Local Data Stewards manually verify these cases to ensure Master Farmer Profile accuracy.

VII. Coclusions and Recommendations

A. Summary of Findings

This study set out to address the Information Siloes experienced in Philippine government agencies most especially for the agricultural sector in Isabela, Philippines. Our analysis confirmed that while the Department of Agriculture (DA), PAGASA, and BSWM produce high-value data, its fragmentation across incompatible systems renders it inaccessible to the smallholder farmer.

The proposed Unified Data Mesh Architecture, built on the Microsoft Fabric platform, demonstrates a technical pathway to resolve this interoperability gap.

- *Ingestion* - The ELT (Extract-Load-Transform) strategy proved superior for AI applications, preserving the raw context of unstructured research documents essential for semantic analysis.
- Virtualization - The use of Fabric Shortcuts validated a Zero-Copy integration model, effectively virtualizing massive BSWM geospatial datasets without the cost or latency of physical replication.
- Governance - The algorithmic Master Data Management (MDM) approach utilizing Fuzzy Logic provided a scalable mechanism for establishing a Single Version of Truth for farmer identities, a prerequisite for hyper-personalized advisory.

B. Key Takeaways

- Integration is the Prerequisite for Intelligence: Advanced AI models like Retrieval-Augmented Generation (RAG) cannot function safely without a governed, integrated data foundation. The Garbage In, Garbage Out principle applies strictly; Project TIWALA's Safety-by -Design framework ensures that AI is grounded in authoritative government data, mitigating the risk of hallucination.
- SaaS Unifies Government IT - The shift from fragmented Best-of-Breed open-source tools to a unified SaaS platform (Microsoft Fabric) offers the Philippine government a viable path to modernization. It simplifies procurement, centralizes security governance (Purview), and reduces the technical debt associated with maintaining complex on-prem infrastructure.
- Resilience Requires Hybrid Architecture - While cloud-native architectures offer scalability, the reality of the Philippine climate demands a hybrid approach. The Last-

Mile delivery system is only as resilient as its edge nodes; thus, local caching mechanisms are non-negotiable for disaster-prone regions like Isabela.

C. Recommendation and Future Work

1. Implementation of Edge Computing Nodes

Future iterations of this architecture should focus on the physical deployment of Edge Computing Hubs in key municipal agriculture offices. These nodes should be capable of running lightweight containerized versions of the inference engine to ensure service continuity during typhoon-induced internet blackouts.

2. Standardization of Government APIs

While Project TIWALA mitigates heterogeneity through ELT, long-term sustainability requires policy intervention. We recommend that the Department of Information and Communications Technology (DICT) mandate the adoption of standard OpenAPI (Swagger) specifications for all agricultural agencies to reduce the engineering overhead of ingestion pipelines.

3. Longitudinal Impact Study

Following the technical deployment, a longitudinal study is recommended to quantify the economic impact of the system. Metrics should focus on the correlation between Time-to-Advice (latency reduction) and Yield Protection (crop loss avoidance) to validate the return on investment for this integration infrastructure.

ACKNOWLEDGMENT

The authors express sincere appreciation to Pablo A. Gumpal, Jr. for providing farming insights and administering the survey, and to Bernadette R. Gumpal for facilitating survey distribution.

The authors also acknowledge the judicious use of generative artificial intelligence (AI) tools, specifically Google's Gemini, Microsoft Copilot, and OpenAI's ChatGPT. These tools were utilized for tasks such as brainstorming ideas, refining language for formality and academic tone, and extracting data for analysis.

All AI-generated outputs were thoroughly reviewed and significantly edited by the authors, who maintain full responsibility for the paper's content and arguments.

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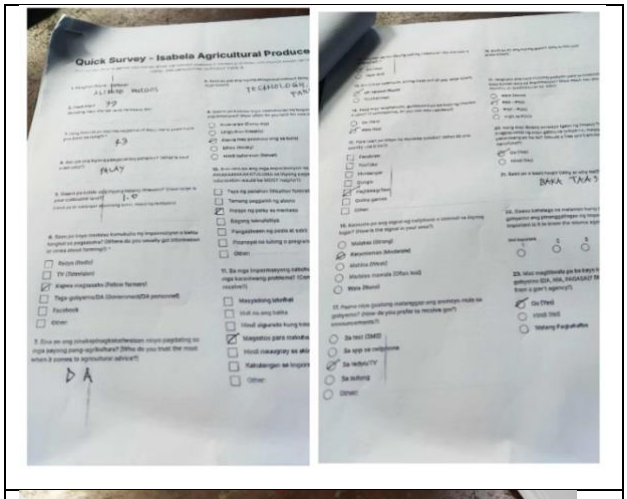
APPENDIX A: INSTRUMENTS & SCREENS

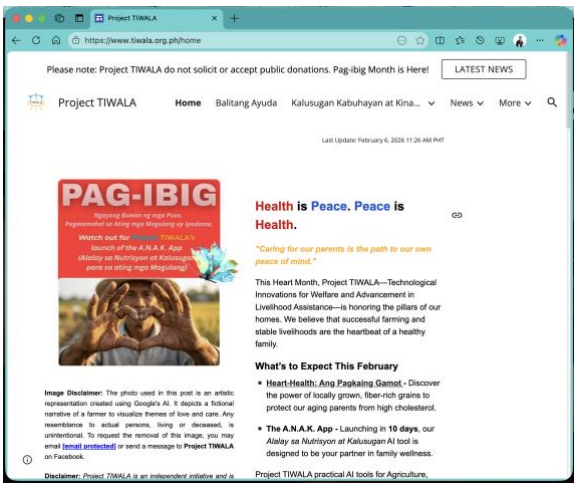
This appendix contains the survey questionnaire distributed to farmers and a selection of high-fidelity prototype screens for the TIWALA application.

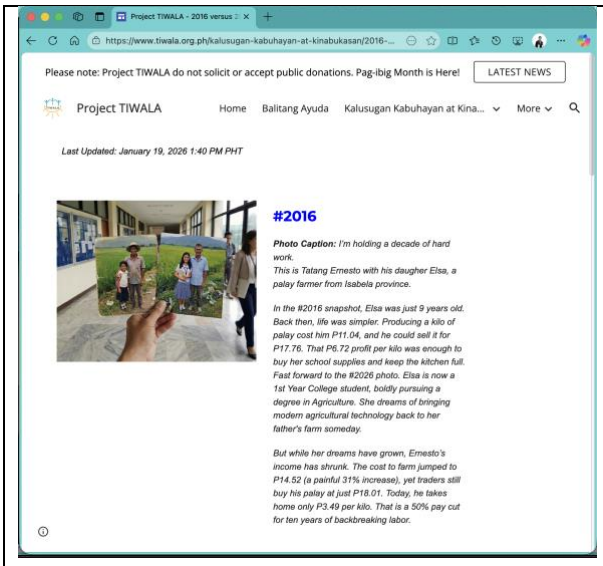
A.1 Original Survey Instrument used in unpublished paper October 2025 "Project TIWALA: A Data-Driven, Socio-Technical Framework for Agricultural Intelligence in Isabela, Philippines"



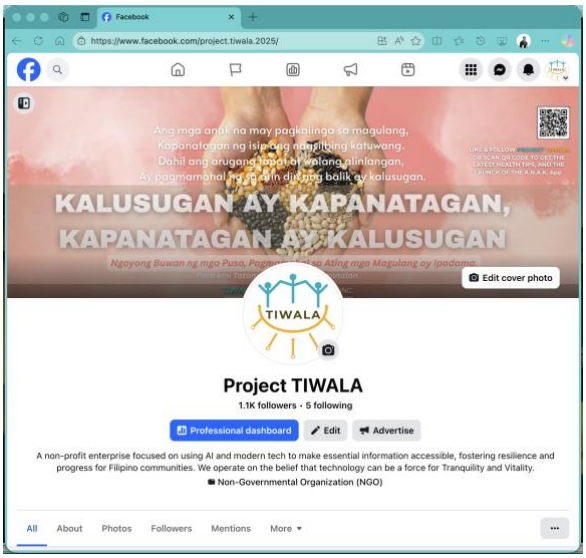
A.2. Project TIWALA Official Page

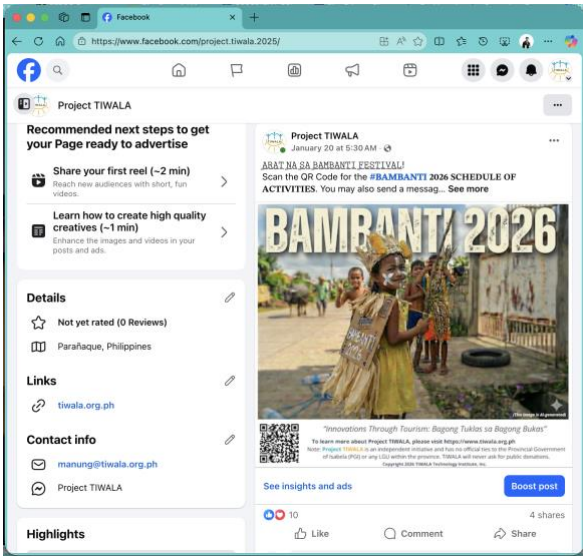
URL: https://www.tiwala.org.ph





Facebook: fb.com/project.tiwala.2025





APPENDIX B: DEFINITION OF TERMS

Technical & Architectural Terms

1. Data Island - Distinct from a "Data Silo". A Data Island refers to a specific, isolated dataset or system that operates independently from the broader enterprise data ecosystem. In the context of this study, the massive geospatial maps managed by the BSWM represent a "Data Island"—valuable and accessible, yet technically disconnected from real-time weather or crop data until bridged by virtualization [12].
2. Data Mesh - A decentralized sociotechnical approach to data architecture where data is treated as a product, owned by domain-specific teams (e.g., DA, PAGASA), rather than a centralized data team. Project TIWALA adopts this by leaving data in its source domain and connecting it via a unified logic layer.
3. ELT (Extract, Load, Transform) - A data integration process where raw data is extracted from the source and loaded directly into the target system (Data Lake) before transformation occurs. This differs from traditional ETL, allowing for the preservation of raw context essential for AI analysis.
4. Medallion Architecture (Bronze / Silver / Gold)

A data design pattern used within the OneLake environment to organize data quality:

- Bronze Layer (Raw) - The landing zone for immutable, raw data ingested directly from sources (e.g., PDF research papers, raw JSON weather logs) without modification.
- Silver Layer (Validated) - The "filtered" zone where data is deduplicated, cleaned, and structurally validated (e.g., converting text dates to standard formats).
- Gold Layer (Curated) - The consumption zone containing highly refined, aggregated data ready for AI synthesis and reporting (e.g., a specific "Daily Advisory" record combining soil + weather + crop stage).

5. OneLake - A unified, logical data lake within the Microsoft Fabric platform that provides a single storage system for all organizational data, supporting "Shortcuts" to virtualize external data without physical movement.
6. Retrieval-Augmented Generation (RAG) - An AI framework that improves the accuracy of Large Language Models (LLMs) by grounding them in external, verified knowledge bases (such as government bulletins) before generating a response, thereby reducing hallucinations.
7. Zero-Copy Integration - An architectural pattern where data is accessed and analyzed directly from its source location without creating physical duplicates. This is achieved in Project TIWALA using Fabric Shortcuts to access external geospatial files.

Agricultural & Domain Terms

1. Climate-Smart Agriculture (CSA) - An integrated approach to managing landscapes—cropland, livestock, forests, and fisheries—that addresses the interlinked challenges of food security and climate change. Project TIWALA serves as a digital enabler for CSA by providing timely weather-contextualized advice.
2. Crop Cycle Phases

The distinct biological stages of plant growth that determine specific resource requirements.

- Vegetative Phase - The period of active leaf and stem growth, requiring high nitrogen inputs.
- Reproductive Phase - The period of flowering and grain formation, where water levels and phosphorus/potassium become critical.
- Ripening Phase - The final stage before harvest where grain hardens and moisture content drops.

3. Registry System for Basic Sectors in Agriculture (RSBSA) - The official national database of farmers, fisherfolk, and farm laborers in the Philippines maintained by the DA.
4. Smallholder Farmer - Farmers cultivating small plots of land (typically less than 2 to 3 hectares in the Philippines), often relying on family labor and facing limited access to capital, technology, and markets.